

192.151 Introduction to Deep Learning

2026S

Project 19: Image Denoising

1. Background and Motivation

Image denoising aims to recover a clean image from a noisy observation, and is a canonical low-level vision problem. It is a building block for many downstream tasks (e.g., segmentation, detection, medical imaging), and it highlights practical issues in deep learning systems such as robustness, generalization to unseen noise, and the trade-off between quality and efficiency. Real-world denoising is additionally challenging, because camera noise is signal-dependent and differs from synthetic Gaussian noise.

2. Problem Description

The primary goal is to build an image denoising system and evaluate it with standard restoration metrics.

a) Dataset and Noise Model (choose one setting):

- **Synthetic noise (recommended):** train on clean images with additive Gaussian noise, e.g., $\sigma \in \{15, 25, 50\}$, and evaluate on standard test sets (e.g., Set12 / BSD68).
- **Real noise (optional):** use a real-noise dataset such as SIDD, and evaluate with the dataset protocol.

b) Denoising Model Implementation:

- CNN/U-Net-like denoiser or DnCNN variant with residual learning.
- NAFNet-style architecture for image restoration.
- (Optional) SwinIR-style restoration model (may require more compute).

c) Evaluation and Analysis:

- Report **PSNR** and **SSIM** on the test set.
- Provide qualitative examples and discuss failure modes (over-smoothing, texture loss, artifacts).
- **Noise-level ablation:** compare performance across multiple σ values.
- **Noise-type generalization:** train on Gaussian noise but test on mild corrupted variants (e.g., Poisson-like noise or JPEG artifacts) and discuss robustness.
- **Efficiency report:** parameter count, peak GPU memory, and inference time per image (or throughput).

3. Expected Deliverables

Students working in groups (ideally 3 members) are expected to:

- produce a project report,
- deliver a group presentation,
- submit well-commented source code (training + inference + evaluation scripts).

4. Suggested Resources and References

- [1] Zhang, K., Zuo, W., Chen, Y., Meng, D., & Zhang, L. (2017). Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising (DnCNN). *IEEE Transactions on Image Processing*. <https://arxiv.org/pdf/1608.03981>
- [2] Liang, J., Cao, J., Sun, G., Zhang, K., Van Gool, L., & Timofte, R. (2021). SwinIR: Image Restoration Using Swin Transformer. In *ICCV Workshops*. <https://arxiv.org/pdf/2108.10257>
- [3] Chen, L., Chu, X., Zhang, X., & Sun, J. (2022). Simple Baselines for Image Restoration (NAFNet). In *ECCV*. <https://arxiv.org/pdf/2204.04676>
- [4] Abdelhamed, A., Lin, S., & Brown, M. S. (2018). A High-Quality Denoising Dataset for Smartphone Cameras (SIDD). In *CVPR*. <https://arxiv.org/pdf/1804.02603>
- [5] KAIR: Image Restoration Toolbox (PyTorch). <https://github.com/cszn/KAIR>
- [6] BasicSR: Open Source Image/Video Restoration Toolbox. <https://github.com/XPixelGroup/BasicSR>